

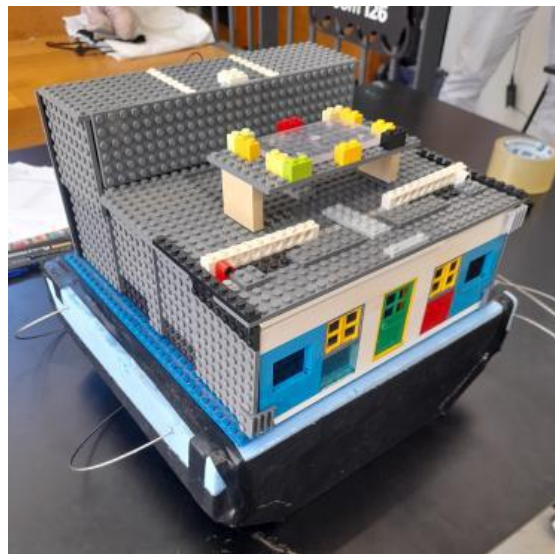
Floor Resistant Houseboat

Brief:

A proof-of-concept prototype for a flood-resistant, station-keeping floating community centre, designed and built as part of a seven-member first-year engineering team. The project was structured across three subsystems, Structure, Drive, and Power & Control, which were developed in parallel with a systems analysis approach to allow independent testing and integration.

The structure consisted of a polyethylene foam flotation base, a plywood foundation, and a LEGO-panelled house selected through rigorous decision matrix analysis weighing structural integrity, water resistance, weight, and sustainability. The drive subsystem used a twin-propeller design with independent DC motors for multidirectional manoeuvrability, supplemented by non-motorised steel cable anchors as an engineered fail-safe for station-keeping. The power and control subsystem used an Arduino UNO microcontroller paired with an HC-05 Bluetooth module and L293D motor driver, controlled via a custom Android app that sent directional commands wirelessly to the motors. The system was successfully tested for buoyancy, wind loading, and stability across multiple prototype iterations.

Final Design:



Reflection:

On demonstration day, the HC-05 Bluetooth module failed to connect due to significant wireless interference in the testing environment, something that had not been encountered in prior testing. This was a valuable lesson in designing for real-world conditions, not just controlled ones. In hindsight, a wired fallback control option or a more interference-resistant communication protocol such as Wi-Fi or an infrared module should have been incorporated as a contingency from the start, rather than relying solely on Bluetooth. This was a valuable lesson in planning ahead and ensuring fail safes are in place before going live.